

COMPUTER SECURITY · FINAL UPDATE · PROJECT FORUM

Phishing Email Detection Using Large Language Models

A leakage-aware, cross-corpus and cost-aware benchmark of small open LLMs

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From a suspicion to a measured result

15 JUL Literature review

5 papers, all 97 to 99%.
Every one tests inside one
dataset. None reports cost.

21 JUL Preliminary run

Train on one corpus, test on
another: F1 0.99 fell to 0.30.

11 AUG Overlap check

Kaggle already contains
two thirds of
SpamAssassin.

5 SEP Proposal talk

Six objectives, one table to
fill.

18 SEP Final experiments

9 sources, 9 models,
164,871 emails checked for
overlap.

What changed since the proposal

Prior work found

E-PhishGen (AISec 2025) already tested
classical models across corpora. So our claim
is now leakage removal plus cost, and we cite
them.

Nazario is test-only

Enron became a full corpus, so pairing Nazario
with Enron mail would leak. Nazario now
measures phishing recall only.

Numbers rebuilt

The 74.4% / 4,282 / 48 on our proposal slides
came from an unsaved run. Every number
today comes from code in the repo.

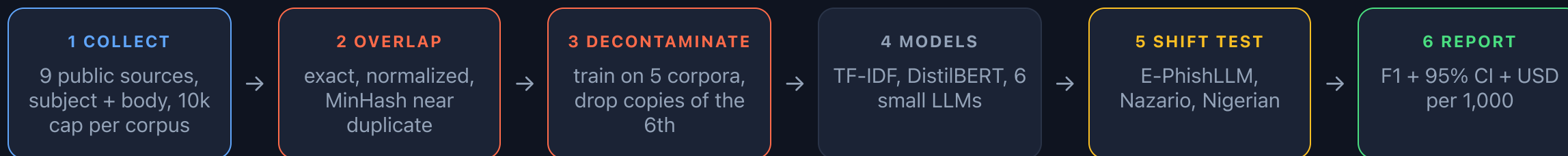
Question: if we remove leaked emails and always test on a corpus the model has never seen, how good are cheap detectors, and what do they cost?

Six objectives from the proposal

#	Objective	What we delivered	Status
1	Decontaminated benchmark	9 sources, overlap measured at 3 levels, near duplicates removed	Done
2	Leave-one-corpus-out	6 folds, each run with raw and cleaned training data	Done
3	One protocol, all families	2 classical, DistilBERT, 6 small LLMs, 2 few-shot. Phishsense-1B not run	Mostly
4	Cost in dollars	Real cost of every one of 19,200 LLM calls, from OpenRouter	Done
5	AI-written phishing	E-PhishLLM (GPT-4o-mini emails) as a held-out shift test	Done
6	One honest table	F1 on unseen corpora with 95% bootstrap intervals, next to cost	Done

Phishsense-1B needs a HuggingFace access token. It is the first item for the next two weeks.

Six steps, one script each, all on a laptop



Leave-one-corpus-out

For each corpus T, train on the other five and test on all of T. Run twice: raw training data, and with every near duplicate of T removed. The gap is the inflation caused by leaked emails.

Same emails for everyone

A fixed 300-email subset per test set. Every model, local or paid, is scored on exactly those emails, so the numbers are comparable.

What exactly we ran

Source	Role	Cleaned	Used	Eval
spamassassin	train+test	5,803	5,803	300
ceas08	train+test	39,091	10,000	300
trec07	train+test	53,638	10,000	300
ling	train+test	2,858	2,858	300
enron	train+test	29,604	10,000	300
kaggle	train+test	17,505	10,000	300
nazario	test only	1,554	1,554	150
nigerian	test only	3,316	3,316	150
ephishllm	test only	11,502	10,000	300

Classical

TF-IDF 1-2 grams, 50k features, LogReg / Naive Bayes, CPU 0.25 s per 1,000 emails

DistilBERT

1 epoch, 128 tokens, batch 16, lr 5e-5, 1,000 emails per corpus, laptop CPU

6 LLMs via OpenRouter

Llama 1B/3B/8B, Qwen 7B, Gemma 12B, Phi-4 14B. Temperature 0, 5 output tokens, 1,500 chars. Few-shot: 4 examples from other corpora

Reproducible

Seed 42 everywhere. 8 GB MacBook. Total API spend \$0.52

The standard check finds almost none of it

SpamAssassin emails that also sit inside the Kaggle set

EXACT TEXT

2

0.03%

NO WHITESPACE

3,850

66.34%

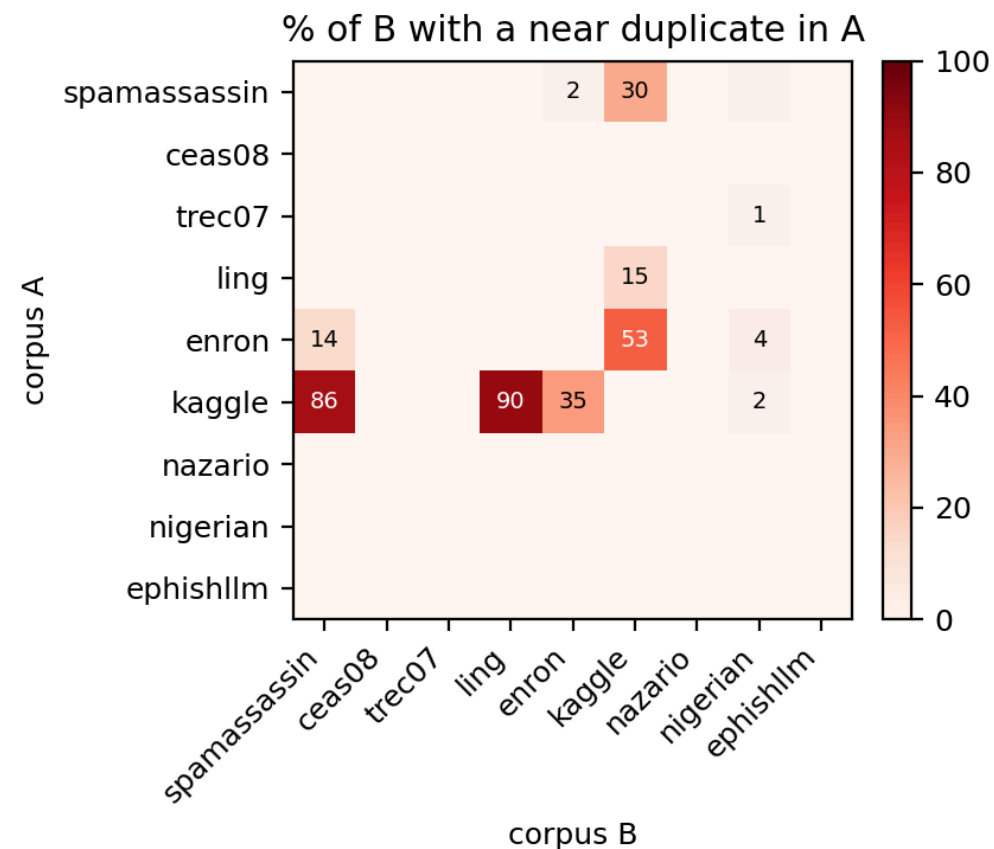
NEAR DUPLICATE

5,009

86.32%

Kaggle also holds **90%** of Ling and **35%** of Enron. More than half of Kaggle is Enron.

Why exact matching fails: Kaggle deletes line breaks and glues words ("cream? Isn't"). We checked 400 of them by brute force: the matches are real.



Remove the copies and the good transfer disappears

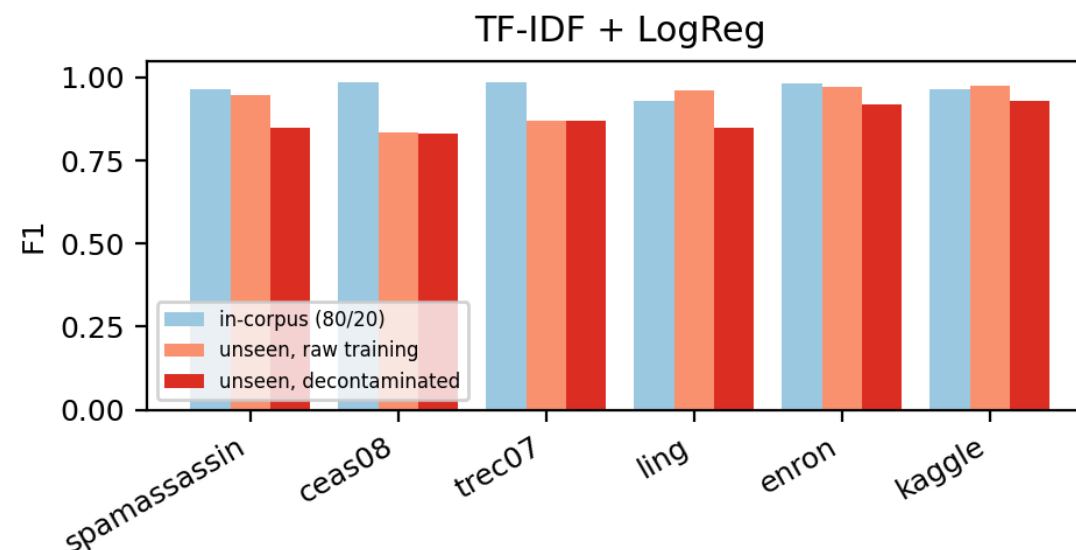
KAGGLE → SPAMASSASSIN, TF-IDF + LOGREG

0.966 → **0.666**

F1 with leaked emails, then after removing them

Held-out corpus	Raw	Clean	Drop	Removed
spamassassin	0.947	0.847	10.0	3,165
ling	0.962	0.847	11.5	1,444
enron	0.972	0.919	5.4	6,091
kaggle	0.976	0.928	4.8	11,018
ceas08	0.833	0.832	0.1	13
trec07	0.871	0.871	0.0	7

CEAS and TREC share almost nothing with the rest, and their scores do not move. That is the control.



Unseen corpora, AI phishing and cost in one place

Model	Unseen-corpus F1 (95% CI)	Worst corpus	False alarms	AI phishing F1	Nazario recall	\$ / 1,000
Qwen-2.5-7B few-shot	0.937 (0.925-0.948)	0.844	5%	0.598	0.847	\$0.074
Qwen-2.5-7B	0.928 (0.915-0.940)	0.850	3%	0.819	0.940	\$0.031
TF-IDF + LogReg	0.901 (0.886-0.916)	0.807	9%	0.640	0.713	local
Llama-3.2-3B few-shot	0.901 (0.885-0.915)	0.848	3%	0.438	0.680	\$0.037
TF-IDF + NB	0.895 (0.879-0.909)	0.773	7%	0.631	0.600	local
DistilBERT	0.891 (0.875-0.906)	0.838	6%	0.493	0.760	local
Gemma-3-12B	0.882 (0.866-0.896)	0.768	25%	0.955	0.980	\$0.016
Llama-3.1-8B	0.881 (0.865-0.896)	0.779	21%	0.817	0.973	\$0.011
Llama-3.2-3B	0.850 (0.832-0.868)	0.799	5%	0.507	0.827	\$0.016
Phi-4-14B	0.727 (0.702-0.749)	0.621	28%	0.759	0.660	\$0.021
Llama-3.2-1B	0.623 (0.597-0.647)	0.359	60%	0.474	0.760	\$0.009

All scores on the same held-out emails. Trained models use decontaminated data. LLMs never saw any of our data. False alarms = share of legitimate mail flagged.

No single winner: pick the trade-off you can live with

1. Best balance: Qwen-2.5-7B zero-shot

0.928 F1 on unseen corpora, 3% false alarms, 0.819 on AI phishing, \$0.031 per 1,000 emails.

2. Gemma-3-12B catches more, flags more

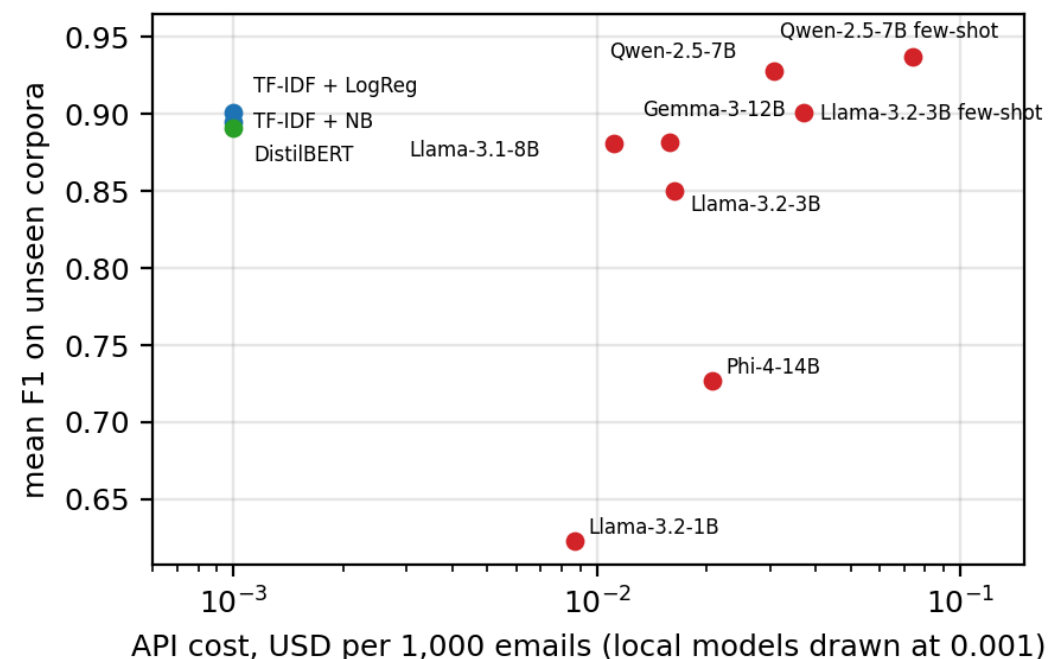
0.955 on AI phishing, but 25% of legitimate mail flagged. Good for review queues, not for blocking.

3. Few-shot helps old mail, hurts AI phishing

Qwen on AI phishing: 0.819 to 0.598 with 4 old examples. They anchor it to old spam.

4. The usual metric picks the wrong model

DistilBERT: best in-corpus F1 (0.975), weakest trained model on AI phishing (0.493).



What this study does not show

Labels are mixed

Four corpora count ordinary spam as positive. We keep the published labels and report phishing-only recall separately.

Small evaluation sets

300 emails per test set keeps the cost low. The 95% intervals are about ± 0.01 to ± 0.03 .

LLM training data is unknown

Old public corpora may be in the LLMs' pre-training data. E-PhishLLM is newer, but we cannot rule it out.

Text only, English only

No headers, URLs or attachments. The Italian and German part of E-PhishLLM is unused.

Light DistilBERT training

1,000 emails per corpus and one epoch, to fit an 8 GB laptop.

One prompt, one run

No prompt search. Providers can change a model behind the same name.

What we can finish before the final paper

DAYS 1 TO 3

Phishsense-1B

Get a HuggingFace token, run it locally on the same 2,400 emails. It is the model that fell from 97.5% to 70%.

DAYS 4 TO 6

Phishing-only labels

Relabel spam vs phishing on a sample and rerun, to close the label gap.

DAYS 7 TO 9

Bigger test sets

1,000 emails per set for the top 3 models. Under \$0.50 more.

DAYS 10 TO 14

Multilingual + write-up

Italian and German E-PhishLLM, stronger DistilBERT, final paper.

Each item reuses the existing scripts. The budget left is well over one dollar.

Everything is public and re-runnable

GITHUB

github.com/Jahid15/phishing-llm-cross-corpus

code, results, paper, research log

COLAB NOTEBOOK

[notebooks/phishing_llm_cross_corpus.ipynb](#)

loads every result in seconds, or re-runs all six steps

High accuracy on one dataset is easy. **Honest evaluation is the hard part.**

Thank you. Questions?